



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Fall 2026)

Machine Learning

Lecture 4: Review of Probability and Statistics

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Today's agenda

- Probability
 - Basic concepts
 - Probability properties
 - Random variable and distribution
 - Expectation and variance
 - Independence
 - Bernoulli distribution and Gaussian distribution
- Statistics
 - Maximum likelihood estimation

Basic concepts

- Experiment:
 - An action or process that leads to one or more possible outcomes.
- Outcome:
 - A single possible result of an experiment.
- Sample space:
 - The set of all possible outcomes of an experiment.
- Event:
 - A subset of the sample space. It is a collection of outcomes that share a common property.

Types of events

- Simple event:
 - An event that consists of exactly one outcome.
- Compound event:
 - An event that consists of more than one outcome.
- Mutually exclusive events:
 - Events that cannot occur simultaneously.
- Independent events:
 - Events where the occurrence of one event does not affect the occurrence of another.
- Complementary events:
 - If event A occurs, then the complement event A' does not occur, and vice versa.

Probability properties

- Non-negativity:
 - $P(A) \geq 0$ for any event A .
- Normalization:
 - $P(S) = 1$ for the whole sample space.
- Joint probability:
 - $P(A, B)$
- Marginal probability:
 - $P(A) = \sum_{b \in B} P(A, B)$
- Conditional probability:
 - $P(A | B) = P(A, B)/P(B)$

Probability of events

- For a finite sample space with equally likely outcomes,
 - $P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes in sample space}}$
 - Example: a fair die with 6 outcomes
- Bayes' Theorem:
 - Find the probability of an event based on prior knowledge of conditions related to the event:
 - $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$

In-class exercise: Bayes' theorem

- $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$
- Suppose you have two coins:
 - Coin A is a fair coin (50% heads, 50% tails).
 - Coin B is biased, with a 70% chance of landing heads and 30% chance of landing tails.
- You randomly choose one of the two coins (with equal probability) and flip it. The result is heads. What is the probability that you chose the biased coin (Coin B)?

Random variable and distribution

- A random variable X is a numerical outcome of a random experiment
- The distribution of a random variable is the collection of possible outcomes along with their probabilities:
 - Discrete: $p(X = x) = p(x)$
 - Continuous: $p(a \leq X \leq b) = \int_a^b p(x) dx$

Expectation

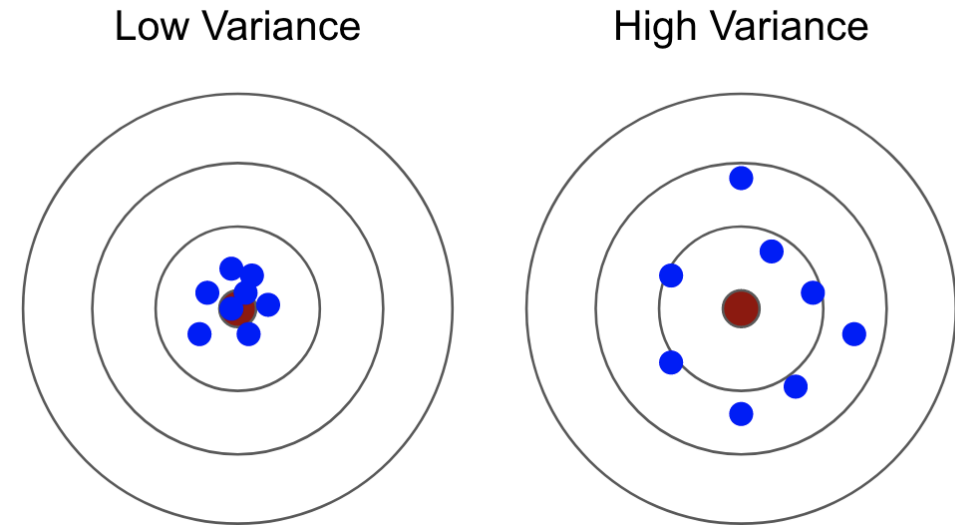
- Discrete case:
 - For a random variable $X \sim p(X = x)$, its expectation is
 - $E[X] = \sum_x xp(X = x)$
 - In an empirical sample, $x_1, x_2, \dots, x_N$, $E[X] = \frac{1}{N} \sum_{i=1}^N x_i$
- Continuous case:
 - $E[X] = \int_{-\infty}^{\infty} xp(x)dx$

Properties of expectation

- Non-negativity:
 - If $X \geq 0$, then $E[X] \geq 0$.
- Linearity:
 - $E[X + Y] = E[X] + E[Y]$
 - $E[aX] = aE[X]$
- Discussion: expectation of $f(x)$, a function of random variable x ?
 - $E[f(x)] = \int f(X)p(x)dx$
 - $E[f(x)] = \sum f(x)p(x)$

Variance

- Variance of a random variable X is the expected value of the squared deviation from the mean:
 - $\text{Var}[X] = E[(X - E[X])^2]$
- Mean $E[X]$
- Deviation $X - E[X]$
- Squared deviation $(X - E[X])^2$



In-class exercise

- Use Markov's inequality to prove Chebyshev's inequality.
 - Markov's inequality:
 - For a *nonnegative* random variable X and any positive number a ,
 - $P(X \geq a) \leq \frac{E[X]}{a}$
 - Chebyshev's inequality:
 - For a *nonnegative* random variable X and any positive number a ,
 - $P(|X - E[X]| \geq a) \leq \frac{\text{Var}[X]}{a^2}$